



Which One Is Wrong?

Quick facilitation notes & answer key

Goal

Students learn the core debugging move for two parallel scripts: compare EACH script to the goal to locate the bug. Here the goal is 'meet on 3'; Bit reaches it and Pixel does not, so Pixel's script is the buggy one. Diagnosing which of two concurrent scripts is wrong (before fixing) is the Grade-2 C3.2 reading-and-locating skill the rest of the subject builds on.

Ontario curriculum (Grade 2 — C3 Coding)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential and concurrent events.

C3.2 — read and alter existing code, including code that involves sequential and concurrent events, and describe how changes to the code affect the outcomes.

Materials

The printed worksheet and a pencil. Optional: a star marker on square 3 (the goal) and two counters; run both and see which counter misses the star.

Run it (about 20 min)

1. I DO: state the goal — 'both should meet on 3.' Run Bit: 0 forward 3 → 3 (on the star).
2. WE DO: run Pixel: 5 back 1 → 4 (misses the star). 'Since Bit is right, the bug is in Pixel.'
3. YOU DO: students name Pixel as the wrong script and say it is 1 away from the goal.
4. Connect: 'Compare each script to the goal — the one that misses is the bug.'

Answer key (modelled by the run-gate)

Ex 1 — Bit: 0 → forward 3 → 3. Pixel: 5 → back 1 → 4. Bit reaches the goal 3; Pixel does NOT (so not both reach it).

Ex 2 — Pixel's script is wrong. We know because Bit already lands on the goal 3, so the bug must be in the other script.

Ex 3 — the goal is 3; the wrong sprite (Pixel) landed on 4, which is 1 number away from the goal.

Diagnosis note: this sheet only LOCATES the bug; fixing specific bug types comes on the next sheets.

Watch for & differentiate

Common error: assuming the bug is in Bit because Pixel 'looks fine' — always check each against the goal number.

Common error: trying to fix before locating — this sheet is about WHICH script is wrong, not the fix yet.

Simplify: mark the goal square, run each sprite, and circle the one that does not land on it.

Extend: ask what single move would fix Pixel (back 2 instead of back 1) — a preview of the next sheets.

Success indicator: the child names Pixel as the buggy script and says it is 1 away from the goal.